

riers, with plans to build more) rapidly increasing its naval footprint in the region, many consider a third carrier is a necessity. With *Vikrant*, India enters an elite group of countries with the capability to design and manufacture indigenous aircraft carriers. The US, UK, Russia, China and France are the other nations to have designed and built aircraft carriers weighing 40,000 tonnes and above.

"She (the IAC) has the soul of the old *Vikrant*. What can be better than having a soul of the ship that showed her mettle in war," Commodore Vidyadhar Harke, the Commanding Officer of the IAC *Vikrant*, tells INDIA TODAY on board the ship. The mammoth vessel is a floating airbase and its flight deck is capable of carrying, arming, deploying and recovering aircraft. The warship can accommodate at least 20 fighter jets and around 10 helicopters—the new MH-60R multirole helicopters, Russian MIG-29K fighter jets and Kamov-31 helicopters for now. The navy has plans to deploy the indigenously built Advanced Light Helicopters (ALH) and Light Combat Aircraft (Naval Version) on *Vikrant*.

The *Vikrant*'s dimensions are in line with other large carriers, but awe-inspiring all the same: 262 metres long and 62 metres at its widest part, it has a height of 59 metres, including the 'tower' or 'island'. There are 14 decks in all, including five in the superstructure. The warship is equivalent to an 18-storey building at its highest point and has over 2,300 compartments designed for a crew of around 1,700 people. It has a top speed of around 28 knots (over 50 kmph) and a cruising speed of 18 knots, with an endurance (the maximum period a vessel can operate unreplenished) of about 7,500 nautical miles. The main engines onboard the ship generate power equivalent to 120 Formula 1 cars and the power used could light up half of Kochi city. The total length of electric cables used for all the systems is around 2,400 km. *Vikrant* has a deep displacement of nearly 45,000 tonnes and is unprecedented in terms of size and complexity in comparison to any ship built in India to date. The vessel is an imposing structure of 21,500 tonnes of special-grade steel developed indigenously and used in Indian naval ships for the first time.



The construction of INS *Vikrant* began in 2009 and the cost came up to Rs 23,000 crore. The Navy opted for CSL due to its bigger dry dock facility and its experience of completing multiple refits of India's other aircraft carriers. During *Vikrant*'s construction, CSL sourced equipment and machinery from major Indian companies like BHEL, BEL, Keltron, Kirloskar, Larsen & Toubro, Wartsila India and over 100 other Indian manufacturers. About 2,000 men worked on the vessel on any given day; over 40,000 were employed indirectly.

The journey to develop the IAC has been full of challenges. Cochin Shipyard had the experience of building 45 commercial ships for European firms, including high-end electric vessels for the oil and gas market. The IAC was its first warship, and standards are much more stringent for a naval ship compared to a commercial one.

CHALLENGES IN CONSTRUCTION

In 2005, when Russia backed out of its promise to supply steel for the carrier, the project was delayed for almost two years. Madhu Nair, chairman-cum-managing director of CSL, says that it was an opportunity for India to develop warship-grade steel. "Defence Metallurgical Research Laboratory (DMRL), a part of DRDO, began working with SAIL to produce warship steel. They came up with special